

State Labor-Market Datasets — Phase-1 Validation Review

Claims · U3 unemployment · Nonfarm payrolls — readiness for lead-lag analysis

TO	Dataset Validation Team	RE	claims / ur / nfp validation decks (Phase-1, modules V1–V4)
FROM	Emmanuel Michaux — Head of Macro Data Science	DATE	9 July 2026
CC	Quant Research	DECISION	HOLD — pass plumbing; not yet cleared for lead-lag

1. Overall assessment

On their own terms these are excellent. The two-clock discipline (observation vs information timestamp), the point-in-time look-ahead guard recomputed independently against the vintage table, the honest treatment of the ALFRED archive boundary, the V3.6 false-alarm analysis (357 phantom "leaks" that are really late *national* archives), and the V4 revision spectrum that organizes the whole trio — claims ~3% revised, U3 86%, NFP 99% — are exactly the right instincts. As a data-integrity pass on each series *in isolation*, I would sign them off.

That is also the limitation. **They validate the plumbing (modules V1–V4) of three series separately. Lead-lag readiness is a different question, and the modules that answer it — V6 cross-sectional coherence, V7 seasonality/stationarity, V8 lead-lag orientation, V9 backtest hygiene — are explicitly deferred to the full suites in every deck.** As presented, Phase 1 clears structure, calendar, point-in-time and revisions. It does not yet clear us to run lead-lag, and I do not want to greenlight on this basis.

2. Critiques that specifically bite lead-lag

NSA vs SA mismatch — the headline gap

Claims are **not** seasonally adjusted; U3 and NFP are. A claims → employment lead-lag on those units will be dominated by the seasonal cycle in claims, not by any genuine lead. This is deferred to V7, but for this goal it is the first thing that will contaminate the result, and nothing in these decks resolves it. Everything must be put on a common footing — YoY log-differences, or seasonally adjust claims — before a single correlation is computed.

No unified cross-series point-in-time panel

We have three PIT panels on two different grids — claims on 818 weekly Fridays, U3/NFP on 239 month-starts. Lead-lag *joins* them, and the join is precisely where look-ahead re-enters. The 0-violation guarantee is per-panel; I have seen no look-ahead guard on the merged object, and the merged object is what a lead-lag study actually consumes.

Weekly → monthly claims alignment is asserted, not shown

The decks correctly note the LAUS/CES reference week is "the week containing the 12th," but leave the aggregation of the ~4–5 weekly claims prints onto the reference month as a rule. For pairing claims with U3/NFP that

construction must be built and validated — respecting both the reference week and the state release availability — or the apparent lead is an artifact of calendar bucketing.

Both sides of the regression must be vintage

With NFP/U3 86–99% revised and the benchmark concentrated in March, regressing a *current* (benchmarked) target on a real-time claims feature manufactures lead-lag out of hindsight. The decks prove the vintage panels exist; they do not commit to vintage-on-vintage *targets*. That commitment needs to be in the protocol.

The claims capture-regime break (V3.7) is a timing hazard, not just a level issue

Pre-2020 the archive stores the revised print at Sat+19d; post-2020 the advance print at Sat+13d, with 1–4 day post-2020 Monday jitter. The ~0.09% level effect is genuinely second-order — but a 6-day availability shift at the era boundary can move a *timing* correlation on the change/momentum features, which is what lead-lag lives on.

Thin turning-point sample

The joint PIT window is bounded by claims at ~2010-11, leaving roughly fifteen years that contain essentially one recession — 2020 — which is a COVID outlier carrying extreme revisions and the capture-regime break inside it. Lead-lag information concentrates at turning points, and in-sample we effectively have one distorted turning point. The confidence intervals will be a 2020 story unless we say so up front.

Governance

The title slides state 25 / 22 / 22 checks, but the scoreboards register fewer because V1 uses "print-style flags" rather than registered rows, so the PASS/WARN totals do not reconcile to the headline. And with 0 FAIL everywhere and every issue a "documented WARN," the gate is effectively self-certifying — no hard blocker is defined. For an auditable gate I want every check registered with an explicit result, and at least one FAIL condition that blocks.

3. Questions for the team

- Where is the merged cross-series PIT panel, and has the look-ahead guard been run on it?
- Show me the exact weekly → monthly claims aggregation used to pair with U3/NFP — does it respect the week-of-the-12th and the state release date?
- Has lead-lag orientation (sign, peak lag) been checked *separately* pre- and post-2020 to rule out the capture-regime break masquerading as signal?
- Will targets as well as features be first-release/vintage? If not, what bounds the revision-driven component of any measured lead?
- Is the design state-to-state, or national-print-conditioned? V3.4 shows the national report leads the state cross-section by ~14 days — that cascade is a cleaner, higher-power signal than state-only.
- cclaims vs iclaims: initial claims run ~7 days fresher — is that being exploited deliberately, or is it a look-ahead trap?
- Was the pit builder run on the *actual* archived release dates through the 2025–26 disruption (median lag 50 → 66d), not the 50-day rule?

4. Suggestions

- Add a short **Phase 1b** covering V6/V7/V8/V9 — that is the real lead-lag gate. As presented, Phase 1 is a plumbing pass.
- Put all three series on a common footing (YoY log-diffs, or seasonally adjust claims) to remove the NSA/SA mismatch before any correlation.
- Build and guard **one** unified as-of panel joining all three series; re-run the 0-look-ahead check on that joined object.
- Pre-register the lead-lag protocol: vintage-on-vintage, estimated per-era (pre/post 2020), 2020-flagged, reported both national-conditioned and state-only, with peak lag, sign and stability.
- Carry the first-print-bias term as an explicit control (U3 first prints overstate by -0.047pp ; NFP understate by $+0.052\%$, sign flipping with the cycle), not a footnote.
- Confirm the one-off data errors (TX $+100\text{k}$; AZ $132 \rightarrow 17,958$) and the Oct-2025 U3 hole are handled in the feature build — winsorized/flagged — not merely documented.
- Define at least one hard FAIL: any look-ahead in the joined panel blocks the stage.

Bottom line. Strong, rigorous plumbing validation — I am glad the vintage discipline is this careful. But I would hold the lead-lag start until I see the joined point-in-time panel, the seasonal alignment across the three series, and the deferred V6–V9 evidence. Reconvene once Phase 1b is drafted.

Sources reviewed: claims_validation_presentation.pdf (39 pp), ur_validation_presentation.pdf (36 pp), nfp_validation_presentation.pdf (36 pp) — Phase-1 validation decks, modules V1–V4.